

THE UNIVERSITY OF HONG KONG

Template for Offering a New Disciplinary Course

Before completing the template, reference should be made to the “Notes to Template for Offering a New Disciplinary Course” (document 254/525) which list the definitions of key terms and explanatory notes of various fields. Modifications to the template to suit disciplinary needs are subject to approval by the Vice-President and Pro-Vice-Chancellor (Teaching & Learning).

I. Course information (Required for approval)

Faculty:	Engineering	Offering Department(s) / School(s)#:	This is a course offered by the Faculty of Engineering
Proposed implementation date and affected cohort(s):	2026/27 Academic Year - As an Engineering core course for the 2026/27 cohort or later. - As a free elective for the 2025/26 cohort or earlier.	Proposed offering semester(s)¹: (please tick as appropriate)	☒ 1 st Semester ☒ 2 nd Semester ☐ Summer ☐ Others: _____
Target student group: (please tick as appropriate)	☒ Ug ☐ TPg ☐ Others: _____	Course type²: (please tick as appropriate)	☒ Disciplinary Course of the Department ☐ Free Elective of <u>Engineering programme</u> (please specify all relevant curricula/programme(s) if it is a cross-listed course) 6353 – Bachelor of Engineering in Civil Engineering [BEng (CivE)] 6987 – Bachelor of Engineering in Computer Engineering / Electrical Engineering / Electronic Engineering [BEng (CE)/BEng (EE)/BEng (ElecE)] 6315 – Bachelor of Engineering in Data and Systems Engineering [BEng (DASE)] 6339 – Bachelor of Engineering in Mechanical Engineering [BEng (ME)]
Course Particulars (as applicable)			
Course code³:	ENGG2202		
Course title:	Engineering Challenges II		
No. of credits⁴:	6		
Pre-requisite(s) & Co-	Pre-requisite: ENGG1101 or ENGG1320		

requisite(s)⁵: (please delete the inappropriate)	Course code(s) and title(s): ENGG1101 Engineering Challenges I or ENGG1320 Engineers in the Modern World Co-requisite: Not applicable Course code(s) and title(s): Not applicable		
Impermissible combination(s)⁶:	Course code(s) and title(s): Not applicable		
Grading system⁷: (please tick as appropriate)	☒ Letter Grades with Grade Points ⁱ (A+ to F) ☐ Pass, Fail or Distinction ⁱⁱ ☐ Pass or Fail ⁱⁱ		
Course Co-ordinator(s) (Please add fields, as necessary)			
Name of Course Coordinator:	Prof. C. K. Chui		
Department:	Faculty of Engineering		
Position:	Principal Lecturer		
Email:	ckchui@hku.hk	Telephone No.:	60815515
Course description⁸ (This will appear on the Syllabus of the curriculum / curricula concerned)			
Each year, the Engineering Faculty presents a set of open-ended, real-world engineering challenges for students to explore. In ENGG2202: Engineering Challenges II, students collaborate in multidisciplinary teams to tackle one of these challenges, engaging in active learning to explore what to learn and how to learn as they navigate engineering problems. This approach develops essential skills for engineers, including adaptability, creativity, and problem-solving, to meet the demands of fast-changing technology. Building on the foundational skills and technical knowledge acquired in ENGG1101: Engineering Challenges I, this course transitions students from structured, guided tasks to tackling open-ended engineering challenges. Student teams are coached by an instructor who facilitates regular active learning activities to support their exploration of the topic, including think-pair-share brainstorming, real-world exploration, prototyping, iterative problem-solving, and peer learning. Through these activities, students develop their technical expertise while enhancing teamwork, communication, and self-directed learning skills.			
Coursework / Examination ratio: <u>100</u> % Coursework, <u>0</u> % Examination			
Course objectives⁹			
This course aims to: • Provide engineering students with hands-on, project-based learning experience. • Develop students' ability to collaboratively tackle real-world engineering challenges by engaging them in projects with multidisciplinary team members and strengthening their practical problem-solving skills. • Promote teamwork and effective communication within diverse, multidisciplinary teams to prepare students for professional engineering environments.			

ⁱ The grade point scale is as follows:

A+: 4.3, A: 4.0, A-: 3.7, B+: 3.3, B: 3.0, B-: 2.7, C+: 2.3, C: 2.0, C-: 1.7, D+: 1.3, D: 1.0, F: 0

ⁱⁱ Approval from the Senate is required for Ug courses to be graded as "Pass, Fail, or Distinction" or "Pass or Fail".

Course learning outcomes¹⁰	
By the end of this course students should be able to demonstrate a threshold level of mastery of the following learning outcomes. The appropriate HKIE abilities associated with each of the learning outcomes are also presented.	
Course Learning Outcomes (CLOs)	Relevant HKIE Abilities for Engineering Graduates (Equivalent to Course Learning Outcomes)
<u>CLO1: Problem-Solving in Engineering</u> Identify and solve engineering problems using appropriate knowledge, skills, and tools, delivering effective solutions within the project scope.	(e) an ability to identify, formulate and solve engineering problems
<u>CLO2: Prototyping and Technical Implementation</u> Prototype, test, and refine engineering solutions using foundational skills in 3D printing, fabrication, electronics, mechatronics, AI, and computer vision.	(c) an ability to design a system, component or process to meet desired needs within realistic constraints, such as economic, environmental, social, political, ethical, health and safety, manufacturability and sustainability
<u>CLO3: Multi-Disciplinary Teamwork and Communication</u> Collaborate effectively in multidisciplinary teams and communicate technical and non-technical information clearly and professionally to diverse audiences, orally and in writing.	(d) an ability to function on multi-disciplinary teams (g) an ability to communicate effectively
<u>CLO4: Professional, Ethical, and Societal Responsibility</u> Demonstrate ethical professionalism and assess solutions for health, safety, sustainability, and broader global/societal impacts.	(f) an ability to understand professional and ethical responsibility (h) an ability to understand the impact of engineering solutions in a global and societal context, especially the importance of health, safety and environmental considerations to both workers and the general public
<u>CLO5: Lifelong Learning and Active Engagement with Contemporary Issues</u> Practice active learning, stay current with emerging engineering trends, and independently acquire new knowledge, skills, and tools for continuous professional growth.	(i) an ability to stay abreast of contemporary issues (j) an ability to recognize the need for, and to engage in life-long learning
Course teaching and learning activities¹¹	
Activities	No. of Hours
Weekly meetings (2-hour active learning activities led by an instructor)	24
Project works	48
Self-study/Readings	24
Assessment (Project reports, presentations)	48
Total:	144

Course assessment methods ¹²			
Assessment method	Type of assessment (e.g. description of assignment)	Weighting (%)	Aligned course learning outcome(s)
Project Progress and Participation (individual)	Checkpoints, written progress summaries and participation in the regular active learning activities (tentatively 12 activities).	24%	CLO 1,3,4,5
Written Reports (individual)	Written reports summarizing the problem definition, analysis, solution, and evaluation.	46%	CLO 1,2,4,5
Solution Demonstration (team)	Presentation or demonstration of a conceptual design or prototype.	20%	CLO 2,3
Teamwork and Collaboration (individual)	Assessment of teamwork, leadership, and contributions throughout the project.	10%	CLO 3
Total		100%	

Note: Individual Faculty may require additional course information for consideration by the Faculty Board / Board of Studies.

II. Other course information (Optional for course approval but is required to be entered into the SIS / Moodle)

Related curriculum / programme requirement ¹³ (please tick as appropriate)						
Curriculum / Programme	Ug				TPg	
Engineering curriculum	☒ Core / Compulsory - As a compulsory core course for the 2026/27 cohort or later. - As a free elective for the 2025/26 cohort or earlier. ☐ Capstone ☐ Disciplinary Elective				☐ Core ☐ Elective ☐ Capstone	

Course learning outcomes and programme learning outcomes alignment ¹⁴						
Curriculum/Programme: _ Bachelor of Engineering (BEng) Degree (refer to document E81/412)						
Course learning outcomes (CLOs)	Aligned programme learning outcomes (PLOs) (Please refer to the PLOs of the respective curricula / programmes.)					
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6
CLO 1	✓	✓				✓
CLO 2	✓	✓				✓
CLO 3					✓	
CLO 4			✓	✓		✓
CLO 5				✓		

(If the course is offered to more than one curriculum / programme, please draw up the matrix table **for each curriculum / programme.**)

Guidelines on grading and standards of assessment¹⁵

(e.g. Course grade descriptors, assessment rubrics, etc.)

Grade Descriptors

Grade	Description
A	Exceptionally good performance demonstrating a superior understanding of the subject matter, a foundation of extensive knowledge, a skillful use of concepts and/or materials, and ability to analyze and evaluate problems.
B	Good performance demonstrating capacity to use the appropriate concepts, a good understanding of the subject matter, and an ability to handle the problems and materials encountered in the course.
C	Adequate performance demonstrating an adequate understanding of the subject matter, an ability to handle relatively simple problems, and adequate preparation for moving on to more advanced work in the field.
D	Minimally acceptable performance demonstrating at least partial familiarity with the subject matter and some capacity to deal with relatively simple problems, but also demonstrating deficiencies serious enough to make it inadvisable to proceed further in the field without additional work.
F	Unacceptable performance demonstrating unfamiliarity with the subject matter, and lack of capacity to deal with relatively simple problems, and also demonstrating deficiencies serious enough to make it advisable to retake the course.

(See attachment I of the assessment rubrics)

Course content and topics¹⁶ (to be entered in the SIS / Moodle)**1. Annual Engineering Challenge Topics**

Each year, the Engineering Faculty presents a set of open-ended, real-world engineering challenges for students to explore. In **ENGG2202: Engineering Challenges II**, students collaborate in multidisciplinary teams consist of 5-6 students to tackle one of these challenges, applying **active learning** to explore what to learn and how to learn as they address complex engineering problems.

While students are expected to have foundational skills in areas such as fabrication, electronics, mechatronics, data analytics, and AI, these challenges also provide opportunities for students to acquire new skills and knowledge as they work together to develop innovative solutions. Examples of potential project topics include:

- **UAV-Based Solutions for Safer and Smarter Cities**
This challenge involves designing unmanned aerial vehicle (UAV) systems to address urban safety and efficiency issues, such as traffic monitoring, disaster response, or infrastructure inspection. Students may explore integrating sensors, real-time data analytics, and autonomous navigation into their UAV designs.
- **Intelligent Systems for Healthy and Resilient Buildings**
This challenge focuses on creating smart building technologies to improve indoor air quality, energy efficiency, and disaster preparedness. Students may develop systems that incorporate environmental sensors, AI-driven data analysis, and IoT connectivity to monitor and optimize building performance.
- **Assistive Engineering Solutions for Mobility and Safety**
This challenge aims to design innovative devices or systems to assist individuals with mobility impairments or enhance safety in daily activities. Students may work on solutions such as

exoskeletons, smart wheelchairs, or wearable safety devices, integrating mechatronics, AI, and user-centered design.

- **Green Innovations for ESG-Driven Sustainable Urban Living**

This challenge addresses sustainability goals by developing technologies for renewable energy, waste management, or resource efficiency in urban settings. Students may create prototypes such as solar-powered systems, smart waste-sorting devices, or water recycling solutions, emphasizing eco-friendly materials and processes.

- **More Topics to be listed**

Additional project topics are introduced and updated annually, with contributions from across all departments, ensuring a wide variety of multidisciplinary challenges.

Through these challenges, students develop essential skills for engineers, including adaptability, creativity, and problem-solving, while working collaboratively to address the demands of fast-changing technology.

2. Active Learning Activities Facilitated by Instructors

Student teams are coached by an instructor who facilitates regular **active learning activities** to support their exploration of the topic, the active learning activities will cover:

- **Problem-Solving Brainstorming (Think-Pair-Share)**

Teams collaboratively analyze specific challenges by brainstorming ideas, sharing individual insights, and developing solutions as a group. This activity promotes active learning, critical thinking, creativity, and teamwork.

- **Real-World Exploration and Engagement**

Students connect classroom knowledge with real-world practices through site visits and/or interactions with industry experts. This activity fosters active learning by helping students contextualize their learning and understand the practical implications of their solutions.

- **Prototyping, Testing, and Iteration**

Students design and build prototypes, test their concepts in real or simulated environments, and refine their solutions through iterative improvements. This hands-on process ensures that students engage in active learning to develop practical engineering skills while improving their designs.

- **Progress Reflection and Feedback**

Teams regularly present their progress, reflect on achievements and challenges, and receive constructive feedback from instructors and peers. This iterative evaluation process fosters adaptability and continuous improvement through active learning.

- **Collaborative Peer Learning**

Students share knowledge within their teams and teach one another, enhancing communication skills, reinforcing understanding, and promoting deeper learning of key concepts through active learning.

3. Project Selection Process

- **Topic Listing**

All project descriptions, including objectives, are published on the course platform (e.g., Moodle/Ed) for students to review.

- **Student Application**

Students submit an application form indicating their top three project preferences and a brief statement of interest, outlining how their skills and background align with their selected topics.

- **Matching Process**

Instructors evaluate student applications based on skills, background, and diversity to form balanced teams. Students are then assigned to projects aligned with their preferences where possible, ensuring a fair and effective distribution of resources and opportunities.

Required / recommended readings and online materials¹⁷ (to be entered in the SIS / Moodle)

- Students will access all required readings, online videos, and exercises through the Learning Management System. Regular engagement with these materials is essential for successful course completion.

Means / processes for student feedback on the course¹⁸

- Student Feedback on Teaching and Learning (SFTL) will be collected, and follow-up actions and responses will be communicated to students.
Course policy¹⁹ (including plagiarism, academic honesty, attendance, etc.)
- Plagiarism is strictly prohibited. - Students must complete all online quizzes and project reports independently. - Participation in online learning activities will be monitored and will contribute to each student's overall assessment.
Other additional course information²⁰ (e.g. course schedule, course quota, etc.)
(Please attach additional pages if necessary)

III. Endorsement and approval from relevant authority

If the new course is offered to current students, the offering unit should ensure that students have been apprised of the syllabus changes and no objection has been raised before submission of this template to the Faculty Board / Board of Studies.

Endorsement from the Department / School# (please add fields, as necessary)
Head of the Department / School# of _____
Remarks: Signature: _____ Name: _____ Date: _____
Approval from the Faculty Board and / or the Board of Studies (please add fields, as necessary)
Chairperson of the Board of the Faculty of _____ / Board of Studies for _____ #
The new course proposal is approved / disapproved / requires revision for further consideration#. Remarks: Signature: _____ Name: _____ Date: _____

Delete as appropriate

June 2015

Amended December 2016
 August 2017
 September 2018
 September 2020
 July 2023
 May 2025

Appendix I. Assessment Rubrics

Project Progress and Participation (24%)

Criteria	Grade A (Excellent)	Grade B (Very Good)	Grade C (Satisfactory)	Grade D (Passing)	Fail (F)
Teamwork and Collaboration	Consistently demonstrates exceptional teamwork by actively contributing ideas, collaborating effectively, and supporting team members.	Demonstrates strong teamwork by contributing ideas and collaborating well, with occasional minor gaps.	Demonstrates adequate teamwork, contributing to team efforts but with limited collaboration or engagement.	Demonstrates minimal teamwork, with inconsistent participation or limited contributions to the team.	Fails to demonstrate teamwork; no meaningful participation or collaboration with the team.
Communication in Meetings	Communicates exceptionally well during meetings, presenting ideas clearly and professionally while engaging constructively with others.	Communicates effectively during meetings, presenting ideas clearly and engaging with others.	Communicates adequately but may lack clarity or professionalism in meetings, with limited engagement.	Communicates minimally during meetings, with unclear or vague participation and limited engagement.	Fails to communicate effectively in meetings; little to no engagement or contribution.
Progress Reporting	Provides detailed, accurate, and professional progress reports that clearly document project developments and outcomes.	Provides clear and accurate progress reports that effectively document project developments.	Provides basic progress reports that document general developments but lack detail or depth.	Provides minimal progress reports that are incomplete or lack clarity, with limited documentation.	Fails to provide progress reports or provides reports that are incomplete or poorly documented.
Global, Societal, and Environmental Awareness	Demonstrates strong awareness of global, societal, and environmental considerations in project tasks, with thoughtful contributions to sustainability.	Demonstrates good awareness of global, societal, and environmental considerations, with adequate contributions to sustainability.	Demonstrates basic awareness of global, societal, and environmental considerations, but contributions may lack depth.	Demonstrates minimal awareness of global, societal, or environmental considerations, with weak contributions.	Fails to demonstrate awareness of global, societal, or environmental considerations in tasks or contributions.
Lifelong Learning and Self-Development	Consistently seeks out new knowledge, skills, or tools to improve performance and demonstrates strong initiative for self-development.	Actively seeks out new knowledge, skills, or tools, with good effort toward self-development.	Occasionally seeks out new knowledge or skills but shows limited initiative toward self-development.	Rarely seeks out new knowledge or skills, showing minimal effort toward self-development.	Fails to seek new knowledge, skills, or tools; no evidence of initiative or effort toward self-development.

Final Report (46%)

Criteria	Grade A (Excellent)	Grade B (Very Good)	Grade C (Satisfactory)	Grade D (Passing)	Fail (F)
Problem Identification and Analysis	Clearly identifies and formulates complex engineering problems, providing detailed and well-supported analysis using appropriate tools and evidence.	Identifies and formulates engineering problems with good analysis and adequate use of tools and evidence.	Identifies problems with basic analysis and limited application of tools or evidence.	Identifies problems but analysis is minimal and lacks sufficient evidence.	Fails to identify or analyze engineering problems; lacks clarity, relevance, or use of tools.
Solution Development and Implementation	Proposes innovative, feasible, and well-justified solutions, effectively applying tools and methodologies.	Proposes feasible and well-justified solutions, using tools and methodologies adequately.	Proposes solutions that are feasible but lack depth or strong justification.	Proposes solutions, but they are underdeveloped or poorly justified with little application of tools.	Fails to propose feasible solutions or provide justification; no meaningful use of tools or methods.

Technical Communication	Communicates exceptionally well in oral and written forms, with clarity, professionalism, and logical structure.	Communicates clearly in oral and written forms, with good professionalism and structure.	Communicates adequately, but presentations lack clarity or professionalism in some parts.	Communicates with minimal clarity or professionalism in oral and written forms.	Fails to communicate technical and non-technical information clearly or professionally.
Report Organization and Quality	Exceptionally well-organized report with logical flow, clear formatting, and no grammatical errors.	Well-organized report with minor errors in formatting or flow.	Adequately organized report but contains several errors in formatting, grammar, or flow.	Report is poorly organized, with significant errors in formatting, grammar, and flow.	Report is disorganized, with pervasive errors that impede understanding.
Ethical Decision-Making	Demonstrates a strong understanding of ethical responsibilities, with well-reasoned and thoughtful ethical considerations.	Demonstrates good understanding of ethical responsibilities, with some ethical considerations included.	Demonstrates basic understanding of ethical responsibilities, with limited or general ethical considerations.	Demonstrates minimal understanding of ethical responsibilities and vague ethical considerations.	Fails to demonstrate understanding of ethical responsibilities or ethical considerations.
Global, Societal, and Environmental Impact	Thoroughly evaluates the global, societal, and environmental impact, addressing health, safety, and sustainability comprehensively.	Evaluates the global, societal, and environmental impact with good attention to health, safety, and sustainability.	Provides a basic evaluation of the global, societal, and environmental impact, with limited details.	Mentions global, societal, or environmental impact, but with minimal evaluation or detail.	Fails to evaluate or address global, societal, or environmental impacts effectively.
Contemporary Issues and Lifelong Learning	Demonstrates excellent awareness of contemporary issues and trends, with significant independent learning and application of new knowledge.	Demonstrates good awareness of contemporary issues and trends, with some evidence of independent learning.	Demonstrates basic awareness of contemporary issues and trends, with limited evidence of independent learning.	Demonstrates minimal awareness of contemporary issues or trends, with little or no evidence of learning.	Fails to demonstrate awareness of contemporary issues or evidence of independent learning.

Solution Demonstration (20%)

Criteria	Grade A (Excellent)	Grade B (Very Good)	Grade C (Satisfactory)	Grade D (Passing)	Fail (F)
Team Collaboration and Communication	Demonstrates outstanding teamwork and synergy, with clear and professional communication tailored to the audience.	Demonstrates strong teamwork and effective communication, with minor areas for improvement.	Demonstrates basic teamwork and communication but lacks consistency or audience focus.	Demonstrates minimal teamwork or communication, with unclear or inconsistent delivery.	Fails to demonstrate teamwork or effective communication; communication is unclear or unprofessional.
Clarity and Effectiveness of Demonstration	Provides a highly engaging and well-structured demonstration that clearly explains the design, prototype, or concept.	Provides a clear and structured demonstration, effectively explaining the design, prototype, or concept.	Provides a demonstration that is somewhat clear, but lacks structure or depth in explanation.	Provides a demonstration, but it is unclear or poorly structured, with limited explanation of the design.	Fails to provide a clear or structured demonstration; the explanation is confusing or incomplete.
Problem-Solving and Design Development	Demonstrates innovative problem-solving through a well-developed design or concept, with strong justification and relevance to the problem.	Demonstrates effective problem-solving with a well-developed design or concept, supported by justification.	Demonstrates basic problem-solving with a workable design or concept, but lacks depth or strong justification.	Demonstrates minimal problem-solving, with a poorly developed or underexplained design or concept.	Fails to demonstrate effective problem-solving or provide a relevant design or concept.
Application of Tools and Engineering Principles	Applies appropriate tools, methods, and engineering principles exceptionally well to create a functional or conceptual design.	Applies appropriate tools, methods, and engineering principles effectively, with minor weaknesses.	Applies basic tools and engineering principles, but usage may lack depth or precision.	Applies tools or principles minimally, with significant gaps in depth, precision, or appropriateness.	Fails to apply appropriate tools, methods, or engineering principles effectively.

Teamwork and Collaboration (10%)

Criteria	Grade A (Excellent)	Grade B (Very Good)	Grade C (Satisfactory)	Grade D (Passing)	Fail (F)
Contribution to Teamwork	Consistently contributes more than their share of work, actively engages in all team tasks, and provides high-quality inputs.	Regularly contributes their share of work effectively and participates actively in team activities.	Completes assigned tasks adequately but contributes minimally to team discussions or problem-solving.	Contributes minimally to team tasks; often relies on others to complete their work.	Fails to contribute meaningfully to team tasks or does not complete assigned work.
Collaboration and Communication	Actively collaborates with team members by providing constructive feedback, communicating clearly, and fostering a positive team environment.	Collaborates effectively with team members, communicates clearly, and contributes to a positive team atmosphere.	Collaborates with team members but may not provide constructive feedback or consistently communicate clearly.	Rarely collaborates or communicates clearly, with limited engagement in team discussions or planning.	Fails to collaborate effectively or communicate with team members, hindering team progress.
Leadership and Initiative	Consistently takes initiative, assumes leadership roles when needed, and drives the team towards achieving goals.	Demonstrates leadership and initiative by guiding the team and ensuring timely progress.	Occasionally takes initiative or leadership roles but often relies on others to lead the team.	Rarely takes initiative or leadership roles, often waiting for others to guide or drive progress.	Fails to demonstrate leadership or initiative; avoids responsibilities and contributes little to team efforts.
Professional Responsibility and Accountability	Always demonstrates professionalism by meeting deadlines, fulfilling commitments, and maintaining a high standard of task completion.	Regularly demonstrates professionalism by meeting deadlines and completing tasks to a good standard.	Demonstrates basic professionalism by completing tasks but may miss deadlines or require reminders.	Shows minimal professionalism; frequently misses deadlines or fails to fully complete tasks.	Fails to demonstrate professionalism or accountability; misses deadlines and fails to complete tasks.